

Last lect?

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Hunds rule

$$|L, S\rangle \text{S max is G.S} \quad (1)$$

$$\text{If } s_1 = s_2 \text{ Lmax is G.S} \quad (2)$$

Given $2p^2$.

$$S = s_1 + s_2 \text{ and we know } |s_1 - s_2| \leq S \leq s_1 + s_2 \quad (3)$$

$$(4)$$

Determining L and S for $2p^2$

We have 3 orbitals corresponding to $m_l \in \{1, 0, -1\}$

$$M_L = \sum_i m_{li} \quad (5)$$

$$M_S = \sum_i m_{si} \quad (6)$$

We have the 15 possible ways single spins and double spins subject to exclusion principle. We can tabulate the values of M_L and M_S . We take them in a grid against each other.